



Social-ecological correlates of children's outdoor playtime and their interaction with gender: a national longitudinal study

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ARTICLE INFO

Keywords:

Outdoor play
Physical activity
Children
Public health
Moderators

ABSTRACT

Background: Outdoor playtime (OP) is consistently associated with higher physical activity in children, but it has declined over the last few decades, underscoring the need to better understand its correlates. Guided by the social-ecological model, we explored the correlates of parent-reported OP in Canadian children.

Methods: In December 2020, we recruited 2291 parents of 7- to 12-year-olds across Canada and followed up every 6 months until June 2022 (4 study waves). We asked parents to report their child's OP on weekdays and weekend days during the previous week. We employed generalized estimating equations to investigate correlates of accumulating ≥ 1 h/day of OP, adjusting for household income and study wave. We tested whether gender moderated each correlate in the multivariable model.

Results: The final multivariable model included 12 significant correlates incorporating four at the individual level (child age, gender, independent mobility, and mobile phone ownership), four at the interpersonal level (parent age and gender, perceived behavioural control for physical activity, and dog ownership), two at the community level (social cohesion and school attendance mode) and two at the built/natural environment level (population density and study wave [likely acting as a proxy for season]). Gender moderated the association of five correlates: child age, independent mobility, social cohesion, school attendance mode, and study wave.

Conclusions: As postulated by the social-ecological model, correlates of OP span multiple levels of influence and interactions between levels are evident. Gender also appears to be an important moderator. Findings can inform future interventions to promote OP.

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<https://doi.org/10.1016/j.envres.2025.123625>

Received 29 August 2025; Received in revised form 2 November 2025; Accepted 23 December 2025

Available online 31 December 2025

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1. Introduction

Consistent evidence shows that children are more physically active when outdoors versus indoors and that greater time spent outdoors is associated with higher overall physical activity (Gray et al., 2015; Larouche et al., 2016; Sallis et al., 2000; Stone and Faulkner, 2014). Outdoor playtime (OP) is also associated with many physical and mental health indicators, including better psychosocial health (Haidt, 2024; Larouche et al., 2016; Piccininni et al., 2018), better management of symptoms of attention deficit hyperactivity disorder (Faber Taylor and Kuo, 2009; Kuo and Faber Taylor, 2004), and reduced risk of myopia (He et al., 2015; Xiong et al., 2017). Exposure to outdoor environments also contributes to place attachment and a sense of belonging (Charles Rodriguez et al., 2023), enhances connection with nature (Michaelson et al., 2020), and can encourage pro-environmental behaviours (Otto and Pensini, 2017).

Despite these benefits, consistent evidence suggests that OP has declined over the last few decades (Bassett et al., 2015; Clements 2004; Mullan 2019), paralleling the declines in active transportation (use of travel modes such as walking and cycling) and independent mobility (IM; i.e., children's freedom to move around in their neighbourhood without adult supervision) (Bassett et al., 2015; Hillman et al., 1990; Fyhri et al., 2011; Larouche, 2018; Shaw et al., 2013). Parental concerns about road and personal safety are often discussed as key contributors to these trends (Carver et al., 2008; Francis et al., 2017; Larouche, 2018; Malone et al., 2007). Children and adolescents may also internalize parental fears and view indoor environments as safer and more comfortable (Crawford et al., 2017; Michaelson et al., 2020). Increased screen time and climate change related challenges (e.g., severe weather events, air pollution) may also contribute to the decrease in OP (Haidt, 2024; Lee et al., 2025; Michaelson et al., 2020).

A better understanding of the correlates of OP is needed to inform the development of effective interventions that may counteract the decline in OP. Recent systematic reviews have summarized the literature on the correlates of OP (Larouche et al., 2023a; Lee et al., 2021). Lee and colleagues (2021) concluded that the correlates of OP spanned multiple levels of influence of the social-ecological model, and that the individual, parental, and social environment levels were particularly salient. Focusing exclusively on longitudinal and intervention studies, Larouche et al. (2023a) noted that higher outdoor time at baseline, exposure to multicomponent interventions, and warmer seasons were associated with more outdoor time, whereas implementation of COVID-19 restrictions and exposure to sun safety interventions discouraging outdoor time at midday were associated with lower outdoor time. However, most other correlates examined were assessed in only one or two studies, precluding conclusions about consistency of evidence and underscoring the need for more longitudinal research. In these systematic reviews, about 45–55 % of studies indicated that girls had lower outdoor time and/or outdoor play (Larouche et al., 2023a; Lee et al., 2021). Given that girls consistently exhibit lower physical levels than boys (Dumith et al., 2011; Guthold et al., 2020; Sallis et al., 2000), there is a need for research examining gender as a potential moderator of the correlates of OP.

In summary, the abovementioned evidence suggests that factors associated with OP include characteristics of the child (e.g., gender and screen time), the parents (e.g., concerns about personal and road safety), the natural environment (e.g., season and climate), and interventions/policies (e.g., sun safety interventions and COVID-19 restrictions). Guided by the social-ecological model (Sallis et al., 2006), we explored the correlates of OP across multiple levels of influence in a national longitudinal study involving Canadian parents of 7- to 12-year-olds. Our study began in the second wave of the COVID-19 pandemic (December 2020) and continued until most public health restrictions were removed (June 2022). We also explored the potential moderating role of child gender.

2. Methods

The Active Transportation and Independent Mobility (ATIM) 2 study was an online longitudinal study that involved 2291 Canadian parents at baseline (Larouche et al., 2023b). Parents were recruited by Léger, a market survey firm that has a panel of >450,000 Canadian adults who volunteer for surveys. Such online panels allow for timely recruitment of large, national samples that have a similar composition with respect to socio-demographic characteristics such as education and household income as the target population (Dubé et al., 2021; Göritz, 2007). Firms like Léger also implement robust quality control procedures (Dubé et al., 2021). Importantly, the use of an online panel circumvented more contact-intensive recruitment during the COVID-19 pandemic. Notwithstanding these advantages, our study participants were drawn from a panel and may not be fully representative of the target population (e.g., there could be self-selection bias and panel conditioning) (Göritz, 2007).

Following baseline data collection (December 2020), participants were invited for follow-ups every six months until June 2022, so they completed up to four surveys. A priori sample size calculations for the ATIM 2 study were initially performed to examine changes in continuous measures of IM and movement behaviours between baseline and the last (18-month) follow-up using the G*Power 3.1.9.4 software. It suggested that a final sample of ~400 would provide a power of 80 % to detect a small-to-medium effect size (Cohen's f^2 of 0.17), using an alpha of 0.05. Based on their experience, Léger estimated that a baseline sample of 2300 would be needed to retain at least 400 participants in the last study wave. However, 708 participants completed the last survey.

Before each survey, participants were screened for the following inclusion criteria: 1) ability to complete the survey in English or French; 2) having a 7- to 12-year-old child; and 3) agreeing to be reinvented for follow-ups. If parents had more than one child in this age range, they were asked to respond for their child whose name came first alphabetically. Because Léger continued to screen for these criteria in the follow-ups, children were excluded after turning 13 years. All study procedures were performed in compliance with relevant laws and institutional guidelines. The project was approved by the University of Lethbridge's Human Participant Research Committee (November 2020; #2020-097). The participants' privacy rights have been observed and parents provided consent electronically after reviewing an information letter. Parents received a \$3 (Canadian) incentive for completing each survey and an additional \$10 if they completed all four surveys. For the data collection dates of the four study waves, the Bank of Canada's (2022) national COVID-19 stringency index (range: 0 to 100) was 64.0, 57.6, 42.6, and 17.0 respectively, indicating that public health restrictions declined over time, especially between the third and fourth waves. Although most policies that directly limited OP were lifted by May 2020 (de Lannoy et al., 2020), many children attended school online or in a hybrid (blended) format (LaBonte et al., 2021), especially at baseline, limiting opportunities for outdoor recess and physical education. This paper was written according to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines (Von Elm et al., 2007) and a checklist is provided in Supplementary File 1.

2.1. Procedures

The full survey is provided in Supplementary File 2. OP was assessed with separate items for weekdays and weekend days during the week preceding the survey. We converted the response options (none; less than 1 h; 1 to less than 2 h; 2 to less than 3 h; 3 to less than 4 h; 4 to less than 5 h; and 5 h or more) into midpoints, using 0.5 h for "less than 1 h" and 6 h for "5 h or more". As per Larouche et al. (2016), we estimated weekly OP with the formula: [(5 * OP on weekdays) + (2 * OP on weekend days)]/7. As the weekly OP variable was not normally distributed, we dichotomized it as <1 h/day vs. ≥1 h/day. This cut-point aligns with a previous study by Stone and Faulkner (2014). We

conducted a separate 1-week test-retest reliability study in November 2020, and the kappa value was 0.44. The modest agreement may be partly attributable to temporal variability in OP, differences in weather, and the fact that provinces were incrementally adding new restrictions to deal with the second wave of COVID-19 at that time (Bank of Canada, 2022), all of which may have led to behaviour change between test and retest assessments.

2.2. Individual factors

Parents reported their child's age (in years), gender (boy; girl; they identify as [specify]; I'd rather not say), length of time the child lived in Canada (2 years or less; 3–5 years; 6 years or more; born in Canada), if they had a diagnosed disability or chronic condition, and if they had a mobile phone. We examined children's IM with Hillman and colleagues' (1990) mobility licenses, assessed with six questions corresponding to children's permission to do the following activities on their own: cross main roads, cycle on main roads, travel home from school, travel to other places within walking distance, use public transport, and go out after dark. Each license was coded as “yes” (1) or “no” (0), then licenses were summed to create an IM index ranging from 0 to 6. A previous study found that this index had good test-retest reliability and convergent validity between child and parent responses in English and French (Larouche et al., 2017).

2.3. Interpersonal factors

We used standard questions from Léger to assess parental employment status and household income (Supplementary File 2). Parents were asked to report their age and gender (man; woman; other [specify]; prefer not to answer), the number of children and adults in the household, the type of home they lived in (low rise apartment/condo; high rise apartment/condo; townhouse; semi-detached house; detached house, other), if they had a dog (yes/no), and the number of motor vehicles that they had (0, 1, 2 or more). We assessed parents' perceived behavioral control to support their child's PA with two items similar to Rhodes et al. (2019). These items assessed to what extent parents perceived that they were capable of – and would have the opportunity to – support their child's PA over the next week on a 5-point scale from strongly disagree to strongly agree. We asked parents to rate their concern about COVID-19 on a 3-point scale (not concerned; somewhat concerned; very concerned). Last, we employed an adapted version of the Tolerance of Risk in Play Scale (TRiPS; Hill and Bundy, 2014) to assess parental risk tolerance, which had excellent internal consistency in our study ($\alpha = 0.90$). We made minor changes to the original Australian version to facilitate comprehension by English- and French-Canadian parents (e.g., we added the term “forest” to the item about playing in the “bush” and we specified in parentheses that playing “chase” is the same as “tag”).

2.4. Community factors

We used subscales about parental concerns with traffic and crime from the Neighbourhood Environment Walkability Scale—Youth (Rosenberg et al., 2009). Their respective α values in the current study were 0.53 and 0.90. We used the 5-item social cohesion subscale from Sampson et al. (1997), which had satisfactory internal consistency in our study ($\alpha = 0.77$). It includes items such as “people around my neighbourhood are willing to help their neighbours” and “people in my neighbourhood can be trusted”. We also asked parents how their child attended school at each wave. At baseline, response options were “in person”, “online”, “blended (a mixture of in person at school and online)”, and “non-applicable”. To be more specific, we divided the fourth option into “school was closed last week” and “home-schooled” for the follow-ups. However, for analyses using data for all time points, we combined these categories as “other”.

2.5. Built and natural environment factors

Parents reported their province or territory of residence. We used a custom Python™ script (Python Software Foundation, Beaverton, OR) to gather data on weather based on the participants' 6-digit postal code and data collection dates. The script extracted data on average minimum and maximum daily temperature and total precipitation for the 7 days before survey completion for the nearest Environment Canada weather station. If weather data were unavailable or marked as unreliable by Environment Canada, the script identified the next closest weather station within a 50-km radius from the postal code and so on. Given that data collection cycles occurred in December and June, differences between cycles may reflect seasonal variations, hence this variable was classified under the natural environment.

We geocoded the six-digit postal code to the geographic centre which allowed us to examine other geographic information systems (GIS) measures of the built environment. To define the home neighbourhood, we applied a Euclidian (i.e., straight line) 1600-m buffer (~1 mile) around the home postal code centroid. As discussed by Duffy et al. (2024), this buffer size is commonly used in research with school-aged children, and it was selected based on preliminary testing and pragmatic considerations (e.g., in rural areas, postal codes cover a larger area, so a 400-m buffer might not have included a participants' actual home location and provided a very limited scope of the home environment). Within this buffer, we examined park density, blue space density, greenspace, population density, and intersection density. Data for park and intersection density were obtained from DMTI spatial's (2022) road file. All ice roads, ferries, railroads, and major highways were excluded as they are not conducive to play. The number of road intersections with at least three or more connections within the buffer was tallied and divided by the size of the buffer. Park density was calculated by dividing the total park area within the buffer by the buffer size. Blue space and population density were obtained from Statistics Canada (2019) data available at the dissemination block level. Blue space density was calculated by dividing the total area occupied by lakes, rivers, and oceans by the buffer size. Population density was calculated for each buffer as the average number of people per squared kilometer for all intersecting blocks within the buffer. Last, greenspace was operationalized as the Normalized Difference in Vegetation Index (NDVI). These data are available from all Canadian postal codes and were obtained from the Canadian Urban Environmental Health Research Consortium. We averaged NDVI values for all postal codes intersecting the buffers.

2.6. Data treatment

To minimize small cell size problems, we restricted our sample to parents aged 20–64 years and parents and children who identified as boy/man or girl/woman like previous surveys on changes in PA associated with COVID-19 (Mitra et al., 2020; Moore et al., 2020). These restrictions led to the removal of 33 participants. Respondents who did not provide a valid postal code ($n = 183$) were excluded from models employing the built environment variables. We recoded household income into four categories (CAD\$39,999 or less; \$40,000–99,999; \$100,000 or more; and prefer not to answer). Keeping the latter category allowed us to increase retention of participants in the analyses. We recoded type of home (detached/semi-detached vs. others), parent occupation (working full-time, homemaker, or others), and classified participants as living in one of five geographical regions, namely Pacific (British Columbia and Yukon), Prairies (Alberta, Saskatchewan, Manitoba, Northwest Territories), Ontario, Quebec, and Atlantic (Newfoundland and Labrador, New Brunswick, Prince Edward Island, Nova Scotia). For the length of time the child lived in Canada, we combined the categories “2 years or less” and “3–5 years” given the small number of recent immigrants. Consistent with Duffy et al. (2024), we dichotomized park and blue space density at the median and classified NDVI, population density, and intersection density into quartiles.

2.7. Statistical analyses

First, we calculated descriptive statistics, including means and standard deviations for continuous variables, and frequencies and percentages for categorical variables. Second, we used chi-square and t-tests to compare baseline characteristics of participants who dropped out with those who completed wave 4. In these analyses, we used

Cohen's *d* and Cramer's *V* as indicators of effect size and interpreted them as per [Cohen \(1988\)](#). Third, we used generalized estimating equations (GEE) to examine bivariate associations between social-ecological variables and the likelihood of reporting ≥ 1 h/day of OP. We selected GEE models given their ability to control for repeated measures while retaining available data from all participants who completed at least one survey in the analyses ([Hanley et al., 2003](#)). In the

Table 1

Descriptive characteristics of the sample at baseline stratified by gender.

Variable	Whole sample (n = 2258)		Girls (n = 1092)		Boys (n = 1166)	
	Frequency (%)	Mean (SD)	Frequency (%)	Mean (SD)	Frequency (%)	Mean (SD)
OP ≥ 1 h/day ^a	1378 (61.0)		631 (57.8)		747 (64.1)	
IM index (number of licenses; range: 0–6)		1.9 (1.7)		1.8 (1.6)		1.9 (1.7)
Child age (years)		9.9 (1.7)		9.9 (1.7)		9.9 (1.7)
Number of adults in household		2.0 (0.7)		2.0 (0.7)		2.0 (0.6)
Number of children in household		2.0 (1.0)		2.0 (1.0)		2.0 (0.9)
TRiPS scale (range: 0–30)		16.2 (6.5)		16.3 (6.5)		16.2 (6.5)
Social cohesion scale (range: 1–5)		3.5 (0.7)		3.5 (0.8)		3.5 (0.7)
Traffic safety scale (range: 1–4)		2.5 (0.6)		2.5 (0.7)		2.5 (0.6)
Crime safety scale ^a (range: 1–4)		2.2 (0.8)		2.2 (0.8)		2.2 (0.8)
Parent gender ^a						
Man	803 (35.6)		364 (33.3)		439 (37.7)	
Woman	1455 (64.4)		728 (66.7)		727 (62.3)	
Household income						
\$39,999 or less	256 (11.3)		124 (11.4)		132 (11.3)	
\$40,000 to \$99,999	1018 (45.1)		486 (44.5)		532 (45.6)	
\$100,000 or more	780 (34.5)		386 (35.3)		394 (33.8)	
Prefer not to respond	204 (9.0)		96 (8.8)		108 (9.3)	
Region						
Pacific	261 (11.6)		123 (11.3)		138 (11.8)	
Prairies	435 (19.3)		218 (20.0)		217 (18.6)	
Ontario	878 (38.9)		432 (39.6)		446 (38.3)	
Quebec	513 (22.7)		249 (22.8)		264 (22.6)	
Atlantic	171 (7.6)		70 (6.4)		101 (8.7)	
School delivery						
In person	1629 (72.1)		774 (70.9)		855 (73.3)	
Blended	118 (5.2)		57 (5.2)		61 (5.2)	
Online	380 (16.8)		198 (18.1)		182 (15.6)	
N/A (e.g., home-schooled)	131 (5.8)		63 (5.8)		68 (5.8)	
Disability or chronic condition ^a						
No	2010 (89.0)		998 (91.4)		1012 (86.8)	
Yes	248 (11.0)		94 (8.6)		154 (13.2)	
Child owns a mobile phone ^a						
No	1518 (67.2)		706 (64.7)		812 (69.6)	
Yes	740 (32.8)		386 (35.3)		354 (30.4)	
Type of home						
Other	597 (26.4)		270 (24.7)		327 (28.0)	
Detached or semi-detached	1661 (73.6)		822 (75.3)		839 (72.0)	
Dog ownership						
Yes	869 (38.5)		418 (38.3)		451 (38.7)	
No	1389 (61.5)		674 (61.7)		715 (61.3)	
Employment (n = 2246)						
Work full-time	1430 (63.7)		689 (63.4)		741 (63.9)	
Homemaker	244 (10.9)		128 (11.8)		116 (10.0)	
Other	572 (25.5)		269 (24.8)		303 (26.1)	
Concerns about COVID-19						
Not concerned	311 (13.8)		155 (14.2)		156 (13.4)	
Somewhat concerned	1226 (54.3)		594 (54.4)		632 (54.2)	
Very concerned	721 (31.9)		343 (31.4)		378 (32.4)	
Time since child lived in Canada						
5 years or less	107 (4.7)		54 (4.9)		53 (4.5)	
6 years or more	297 (13.2)		145 (13.3)		152 (13.0)	
Born in Canada	1854 (82.1)		893 (81.8)		961 (82.4)	
Vehicle ownership ^a						
No	136 (6.0)		55 (5.0)		81 (6.9)	
One	909 (40.3)		407 (37.3)		502 (43.1)	
Two or more	1213 (53.7)		630 (57.7)		583 (50.0)	
Parent travel mode to work						
Active	293 (13.0)		139 (12.7)		154 (13.2)	
Motorized	1404 (62.2)		685 (62.7)		719 (61.7)	
N/A	561 (24.8)		268 (24.5)		293 (25.1)	

Note: Descriptive statistics were performed with children and parents identifying as boy/man or girl/woman and parents aged 20–64 years (n = 2258). IM: independent mobility; OP: outdoor playtime; TRiPS: tolerance to risk in play scale.

^a denotes statistically significant differences between boys and girls ($p < 0.05$).

GEE models, repeated measures were nested within participants, and an autoregressive working correlation matrix was specified. Fourth, we developed a multivariable model of the correlates of OP using a backward selection process. We initially retained variables associated with OP at a liberal $p < 0.20$ threshold for potential inclusion in multivariable models (Gropp et al., 2012). Then, we removed non-significant variables ($p > 0.05$) one by one, unless their removal led to a significantly worse model fit as assessed with the corrected quasi-likelihood under independence model criterion (QICC) statistic. We retained household income as fixed effect in the model regardless of significance to control for socio-economic status, which is often associated with PA (Sterdt et al., 2014) and could affect participation in some outdoor activities. Finally, we examined child gender as a potential moderator of the correlates of OP from the multivariable model by fitting interaction terms (Rich-Edwards et al., 2018). In the multivariable models, there was no evidence of multicollinearity (all variance inflation factor values < 1.6). Last, given the finding of significant interaction terms, we computed gender-stratified multivariable models. We performed all analyses in IBM SPSS, version 30 (Armonk, NY).

3. Results

3.1. Descriptive and bivariate statistics

Table 1 provides descriptive statistics of the sample at baseline and identifies gender differences. On average, children (aged 9.9 ± 1.7 years, 51.6 % boys) played outdoors 1.5 ± 1.1 h/day. When OP was dichotomized, 61.0 % of participants played outdoors ≥ 1 h/day at baseline, and the proportion was higher in boys versus girls. More boys than girls had a disability or chronic condition while more girls had a mobile phone. A higher percentage of parents who answered the survey for a boy identified as man. Conversely, parents of girls had slightly higher crime safety concern scores and were more likely to have at least 2 vehicles. At each successive wave, 61.0 %, 81.2 %, 60.1 %, and 76.8 % of children played outdoors ≥ 1 h/day respectively ($\chi^2_{[3]} = 210.573$; $p < 0.001$; Cramer's $V = 0.201$). Results of the bivariate models for each potential correlate of OP are available in Supplementary Table 1, and for clarity and conciseness, we only present findings from the multivariable model (Table 2) by level of influence in sections 3.3. to 3.6.

3.2. Attrition analyses

After exclusions for small cell size, sample sizes at each wave (1–4, respectively) were 2,258, 1,343, 943, and 694 (Supplementary Fig. 1). Follow-up length ranged from 0 to 18 months with an average of 7.9 months. Results of comparisons of baseline characteristics between dropouts and completers are presented in Supplementary Table 2. Compared to completers, parents who dropped out were significantly younger (40.6 vs. 42.4 years), more likely to identify as women (71.0 vs 65.6 %), more likely to have no vehicles (6.9 vs. 4.2 %), to be in the “other” employment category (26.9 vs. 22.3 %), to have a dog (39.9 % vs. 35.4 %), and to report that their child lived in Canada for ≤ 5 years (5.3 vs. 3.4 %). They also perceived lower social cohesion (3.5 vs. 3.6), had higher risk tolerance scores (16.5 vs. 15.6) and reported that their child had a higher IM index (1.9 vs. 1.7. Except for parent age, effect sizes for all these differences were trivial.

3.3. Individual level

Odds of playing outdoors ≥ 1 h/day decreased with each year of age (OR = 0.81; 95 % CI: 0.76–0.86), while they increased for each IM license (OR = 1.13; 95 % CI: 1.07–1.18). Odds were lower among children who had a mobile phone (OR = 0.84; 95 % CI: 0.71–0.99), and in girls versus boys (OR = 0.77; 95 % CI: 0.66–0.90).

Table 2

Multivariable models of the correlates of playing outdoors at least 1 h per day by levels of influence of the social-ecological model.

Variable	OR	95 % CI	p
Individual level			
Child gender – girl (ref: boy)	0.77	0.66–0.90	<0.001
Child age (each additional year)	0.81	0.76–0.86	<0.001
Child has a mobile phone – no (ref: yes)	0.84	0.71–0.99	0.036
Independent mobility index (each unit increase)	1.13	1.07–1.18	<0.001
Interpersonal level			
Parent gender	1.24	1.05–1.47	0.012
Parent age (each additional year)	0.98	0.97–0.99	<0.001
Income – \$40,000–99,999 (ref: \$39,999 or less)	1.18	0.90–1.54	0.229
Income – \$100,000 or more (ref: \$39,999 or less)	1.01	0.77–1.34	0.928
Income – prefer not to answer (ref: \$39,999 or less)	1.10	0.77–1.58	0.608
Dog ownership – yes (ref: no)	1.37	1.17–1.61	<0.001
Perceived behavioural control for PA (each unit increase)	1.25	1.16–1.34	<0.001
Community level			
Social cohesion (each unit increase)	1.14	1.02–1.26	0.016
School attendance mode – online (ref: in person)	0.45	0.37–0.54	<0.001
School attendance mode – blended (ref: in person)	0.70	0.52–0.95	0.021
School attendance mode – other (ref: in person)	0.67	0.49–0.91	0.010
Built and natural environment			
Wave 2 (ref: wave 1) ^a	4.44	3.71–5.30	<0.001
Wave 3 (ref: wave 1)	1.08	0.92–1.27	0.335
Wave 4 (ref: wave 1)	2.70	2.22–3.29	<0.001
Population density – 2083–3235 (ref: <2083)	0.75	0.59–0.95	0.015
Population density – 3235–5048 (ref: <2083)	0.66	0.52–0.84	<0.001
Population density – >5048 (ref: <2083)	0.67	0.53–0.85	<0.001

Note: CI: confidence interval; OR: odds ratios; PA: physical activity.

^a Study wave was categorized under the natural environment because waves 1 and 3 data were collected in December and waves 2 and 4 in June, so this variable likely acted as a proxy for season. Model fit: QIC = 5441.089; QICC = 5431.075.

3.4. Interpersonal level

In the multivariable model, the odds of playing outdoors ≥ 1 h/day decreased with each year of parent age (OR = 0.98; 95 % CI: 0.97–0.99) and were higher among children whose family owned a dog (OR = 1.37; 95 % CI: 1.17–1.61). Women respondents were more likely than men to report that their child played outdoors ≥ 1 h/day (OR = 1.24; 95 % CI: 1.05–1.47). Each unit increase in the perceived behavioral control scale was associated with 25 % higher odds of reporting ≥ 1 h/day of OP (OR = 1.25; 95 % CI: 1.16–1.34). Household income was not associated with OP.

3.5. Community level

Each unit increase in parent-perceived social cohesion was associated with 14 % higher odds of playing outdoors ≥ 1 h/day (OR = 1.14; 95 % CI: 1.02–1.26). Compared to children attending school in-person, children attending in an online (OR = 0.45; 95 % CI: 0.37–0.54), blended (OR = 0.70; 95 % CI: 0.52–0.95), or other format (OR = 0.67; 95 % CI: 0.49–0.91) were all less likely to achieve this benchmark.

3.6. Built and natural environment level

Higher population density was associated with lower odds of playing outdoors ≥ 1 h/day (p-trend = 0.003). Children living in the second to fourth quartile for population density were ~25–33 % less likely to achieve this benchmark compared to those living in the least dense neighbourhoods. Seasonal differences in OP were evident as children were more likely to play outdoors ≥ 1 h/day during waves 2 (OR = 4.44; 95 % CI: 3.71–5.30) and 4 (OR = 2.70 95 % CI: 2.22–3.29), which corresponded to June data collection periods, compared to wave 1 (December).

3.7. Interactions with gender

Significant interaction terms were observed between gender and five correlates of OP from the multivariable model (Table 2): child age, IM index, social cohesion, study wave, and school attendance mode. Boys were treated as the reference group for all interaction tests. Results indicated that the age-related decline in OP was steeper in girls (OR = 0.90; 95 % CI:0.81–0.99). The association between IM and OP was weaker among girls (OR = 0.92; 95 % CI:0.84–1.00) while the reverse was found for social cohesion (OR = 1.25; 95 % CI:1.02–1.52). The increase in OP in wave 4 versus baseline was weaker in girls (OR = 0.62; 95 % CI:0.42–0.90). Last, the negative association between other school attendance format and OP versus in-person attendance was stronger in girls (OR = 0.46; 95 % CI:0.24–0.87). The association between attending school in an online or blended format (vs. in person) and OP did not differ significantly between genders.

3.8. Gender-stratified models

Findings of multivariable gender-stratified models are provided in Table 3. In general, the main findings were similar as child age, parent age, perceived behavioral control, online school attendance, study wave, and population density were consistently associated with OP for both genders. However, IM index and dog ownership were significantly associated with OP only in boys, whereas school travel mode and social cohesion were significantly associated only in girls.

4. Discussion

The primary objective of our study was to explore the correlates of OP among Canadian children over a 1.5-year period during the COVID-19 pandemic based on the social-ecological model. Our multivariable model indicated that significant correlates span multiple levels of influence, including individual (child age, gender, mobile phone ownership, and IM), interpersonal (parent age and gender, perceived behavioral control, dog ownership), community (social cohesion and school attendance mode), and natural and built environment features (study wave and population density). We also found that child gender

moderated the association of five of the above correlates. Notably, both community level correlates were moderated by gender, indicating interactions between levels of influence as postulated by the social-ecological model (Sallis et al., 2006), thus supporting its theoretical relevance for investigating correlates of OP, and in turn for informing policies and interventions.

4.1. Individual level

We found that the odds of playing outdoors ≥ 1 h/day decreased by 19 % with each year of age. In their systematic review, Larouche et al. (2023) noted a curvilinear association between age and outdoor time, with studies typically showing increases in early childhood followed by decreases in late childhood and adolescence. Moreover, previous reviews have repeatedly shown age-related declines in PA among children and adolescents (Dumith et al., 2011; Reilly, 2016; Sallis, 2000). Thus, our results underscore a need for intervention(s) to promote OP and encourage maintenance of outdoor activities with age.

Our multivariable model indicated that girls were 23 % less likely to play outdoors ≥ 1 h/day than boys. Moreover, significant interaction terms indicated that the age-related decline was steeper in girls and that the increase in odds of playing outdoors ≥ 1 h/day at wave 4 (June 2022) versus wave 1 (December 2020) was smaller in girls than boys. While two systematic reviews reported mixed evidence for associations between gender and outdoor time and/or play (Larouche et al., 2023a; Lee et al., 2021), our findings align with a meta-analysis reporting that the age-related decline in PA is steeper in girls (Dumith et al., 2011). Thus, additional gender-tailored efforts to encourage girls to play outdoors may be needed.

Each unit increase in the IM index was associated with 13 % higher odds of playing outdoors ≥ 1 h/day. Our findings concur with previous research indicating that British children with greater IM played outdoors more, were more likely to engage in active transportation and more active overall (Page et al., 2009, 2010). Collectively, these findings underscore a need for interventions to support children's IM as a potential enabler of OP.

Children who had a mobile phone were 16 % less likely to play outdoors ≥ 1 h/day. These findings support recent calls for action to

Table 3
Gender-stratified multivariable models of the correlates of playing outdoors at least 1 h per day.

Variable	Boys			Girls		
	OR	95 % CI	p	OR	95 % CI	p
Individual level						
Child age (each additional year)	0.82	0.77–0.90	<0.001	0.76	0.70–0.83	<0.001
IM index (each additional license)	1.13	1.06–1.21	<0.001	1.07	0.99–1.15	0.093
Child travel mode PM - active (ref: motorized)	–	–	–	1.43	1.12–1.81	0.004
Interpersonal level						
Dog ownership – yes (ref: no)	1.47	1.17–1.85	<0.001	–	–	–
Parent age (each additional year)	0.98	0.97–1.00	0.006	0.97	0.95–0.99	<0.001
Parent PBC scale (each unit increase)	1.30	1.18–1.43	<0.001	1.20	1.08–1.33	<0.001
Income - \$40,000–99,999 (ref: \$39,999 or less)	0.96	0.67–1.40	0.843	1.31	0.85–2.03	0.221
Income - \$100,000 or more (ref: \$39,999 or less)	1.07	0.73–1.57	0.716	0.86	0.55–1.36	0.517
Income - prefer not to answer (ref: \$39,999 or less)	1.23	0.75–2.04	0.413	0.90	0.49–1.64	0.722
Community level						
Social cohesion scale (each unit increase)	–	–	–	1.25	1.07–1.46	0.005
School mode - online (ref: in person)	0.53	0.40–0.69	<0.001	0.37	0.27–0.51	<0.001
School mode - blended (ref: in person)	0.77	0.50–1.19	0.245	0.66	0.42–1.04	0.072
School mode - other (ref: in person)	0.95	0.60–1.51	0.836	0.52	0.31–0.86	0.011
Built and natural environment level						
Study wave – 2 (ref: 1)	3.81	2.98–4.87	<0.001	4.67	3.51–6.23	<0.001
Study wave – 3 (ref: 1)	1.12	0.90–1.41	0.312	1.09	0.86–1.40	0.469
Study wave – 4 (ref: 1)	3.35	2.49–4.52	<0.001	2.11	1.59–2.81	<0.001
Population density – 2083–3235 (ref: <2083)	0.69	0.49–0.96	0.026	0.76	0.53–1.07	0.119
Population density – 3235–5048 (ref: <2083)	0.55	0.39–0.77	<0.001	0.66	0.47–0.93	0.016
Population density – >5048 (ref: <2083)	0.63	0.46–0.88	0.006	0.58	0.41–0.82	0.002

Note: CI: confidence interval; IM: independent mobility; OR: odds ratio; PBC: perceived behavioral control. IM index was retained in the model for girls even if not statistically significant because its removal led to worse model fit. Model fit (boys): QIC = 2758.573; QICC = 2750.920. Model fit (girls): QIC = 2338.364; QICC = 2331.888.

reduce mobile phone use among children in schools and other settings (Haidt, 2024). Parents often give a mobile phone to their child because of safety and/or pressure from the child, but this stimulates screen time and can delay the development of IM (Haidt, 2024; Malone, 2007). A population-based analysis of Canadian data from the Health Behaviour in School-aged Children survey indicated that adolescents' screen-based activities increased with age while connection to nature decreased (Michaelson et al., 2020). Our findings suggest that the substitution of OP by mobile phone use may begin well before adolescence.

4.2. Interpersonal level

Parents' perceived behavioral control to support their child's PA was associated with higher odds of OP. This observation is consistent with previous research indicating that perceived behavioral control is associated with higher PA or OP among children (Moore et al., 2020; Rhodes et al., 2015, 2019). Because we assessed perceived behavioral control in relation to PA rather than OP, we may have underestimated the strength of association with OP. Nonetheless, OP and PA are consistently associated with one another (Gray et al., 2015) and OP is often assessed as a surrogate of PA, especially in the early years (Burdette et al., 2004). Our findings underscore the key role of parents in supporting their children's OP, for example by playing outdoors with them, encouraging them to go outdoors, and providing equipment for activities. Given that the items used to assess perceived behavioral control in our study were not specific to OP, we cannot draw definitive conclusions about the mechanisms explaining its association with OP, emphasizing a need for further research. Based on our findings and previous studies (Chase et al., 2022; Moore et al., 2020; Riazzi et al., 2019), dog ownership may also support OP, IM, and PA among children. Conversely, we observed that children of older parents were less likely to play outdoors ≥ 1 h/day, a finding that is consistent with a national cross-sectional survey of Canadian parents in the first wave of COVID-19 (Moore et al., 2020). Additional effort may be needed to encourage older parents to support their children's outdoor activities, for example, through verbal encouragement and/or co-participation.

We found no associations between indicators of socio-economic status – including household income, parental occupation, and vehicle ownership – with OP, even in bivariate models. In the systematic review by Lee et al. (2021), 69 % of studies showed no association between household income or socio-economic status and children's outdoor play. These results are encouraging from a social determinants of health perspective given that lower household income is often associated with lower PA in children and adolescents (Sterdt et al., 2014). These differences may be related to the affordability of OP relative to other types of PA (e.g., organized sports).

4.3. Community level

Children attending school online during the pandemic were 55 % less likely to play outdoors ≥ 1 h/day than those attending in-person. A similar pattern was found when comparing blended versus in-person formats, though the effect size was smaller. Our findings are consistent with previous studies conducted during the pandemic (Moore et al., 2020; Paterson et al., 2021). In addition, a systematic review found consistent evidence of lower OP associated with COVID-19 restrictions (Larouche et al., 2023a). While the moderation analysis only indicated a significant difference for the comparison between other attendance mode and in-person attendance, the stratified models (Table 3) show similar trends for online and blended formats.

Interestingly, we found no associations between parental concerns about traffic and crime safety and the odds of playing outdoors ≥ 1 h/day in multivariable models, though crime safety was a significant correlate in bivariate models. As many parents worked from home during the pandemic, there would have been a substantial reduction in traffic volume, which may have helped alleviate parental concerns (Mitra

et al., 2020). Moreover, these concerns may be more salient for OP in certain contexts (e.g., playing in the front yard or street) than others (e.g., playing in the backyard or under adult supervision). We found that each unit increase in the social cohesion scale was associated with 14 % higher odds of playing outdoors ≥ 1 h/day, and the association of social cohesion with OP was stronger for girls than boys. The stronger association between social cohesion and OP in girls might stem from gender differences in socialization whereby girls may be raised to rely more on their community whereas boys may be encouraged to be more socially independent (Brown et al., 2008). However, a systematic review examining parental correlates of OP found no significant gender difference in the association between social cohesion and OP (Boxberger and Reimers, 2019), suggesting a need for more research. Hohl et al. (2019) discussed how place-based interventions focusing on remediating vacant land and abandoned buildings can reduce violence and, in theory, reestablish social control and cohesion in neighbourhoods. Yet, to our knowledge, such interventions have not been tested in relation to children's OP and related outcomes.

4.4. Natural and built environment level

We found that children were 2.7–4.4 times more likely to play outdoors ≥ 1 h/day during warmer data collection periods (June vs. December). These findings are consistent with previous longitudinal studies (Larouche et al., 2023a). The inclusion of study wave in the model eliminated the significant associations between warmer minimum and maximum average temperature during the survey recall period and OP in bivariate models (both $p < 0.001$). Thus, our findings emphasize the need for promoting OP in colder seasons. It may also be worth providing advice about how to dress up for winter activities to less acculturated individuals who were found to spend less time outdoors in a Canadian multi-site study (Nayakarathna et al., 2022).

We found that children living in denser neighbourhoods played outdoors less. These findings are consistent with a national survey of Canadian parents conducted in the first wave of COVID-19 indicating that low dwelling density was associated with higher odds of a parent-reported increase in outdoor activities since the beginning of the pandemic (Mitra et al., 2020). These findings could reflect that less dense areas afford more space for children to play, and they may be associated with lower traffic and parental concerns about road safety. Moreover, Janssen & King (2015) previously observed that walkable school neighbourhoods are not necessarily playable neighbourhoods based on Canadian data from the Health Behaviour in School-aged Children survey. While dense and walkable neighbourhoods can facilitate active transportation (especially among adolescents and adults) and help reduce greenhouse gas emissions (Giles-Corti et al., 2016), it is important to ensure that they remain safe and retain child-friendly features that encourage outdoor activities (e.g., parks and playgrounds).

While most other built environment variables were significantly associated with OP in bivariate models, their association became non-significant in multivariable models. These findings suggest that the associations between these variables and OP are not independent. Hence, future research should examine the clustering of built environment features in relation to OP. Currently, the most widely used composite measures of the built environment are walkability indices focusing on characteristics such as land use mix, street connectivity, and population density (Shashank and Schuurman, 2019). Though widely available in geographic information systems datasets, these variables may not correspond to child-friendly built environments (Janssen and King, 2015), emphasizing a need for the development of more relevant composite measures.

Regional differences were only found in the bivariate models (indicating higher OP among children from the Atlantic vs. Pacific region). Regional differences were more important in the first wave of COVID-19 when policies facilitating and restricting access to the outdoors varied by province, and parents in Ontario reported a larger decline in OP

compared to other provinces (de Lannoy et al., 2020; Moore et al., 2020). Policies restricting outdoor activities were largely lifted by the end of May 2020 (de Lannoy et al., 2020).

4.5. Interactions with gender

Social-ecological models posit that behaviors are influenced by multiple levels of influence and that there can be interactions between these levels (Sallis et al., 2006). In the present study, we tested for interactions between gender and all variables included in the multivariable model (Table 2). Moderation analyses indicated that the positive association between IM and OP was weaker in girls versus boys. Gender-stratified models indicated that the odds ratio for each additional mobility license was 1.13 for boys and 1.07 for girls (Table 3), so despite the significant interaction, promoting IM appears beneficial for both genders. As previously mentioned, gender also moderated the associations of child age, study wave, and social cohesion, and school attendance mode with OP. In comparison, the five studies that assessed sex or gender as a moderator in the most recent systematic review of longitudinal studies on outdoor time reported mixed findings (Larouche et al., 2023a), suggesting a need for more robust research in this area. To more precisely test the social-ecological model, researchers should recruit larger samples sizes with the goal of directly testing the complex and likely non-linear nature of the associations between levels of influence.

4.6. Limitations and strengths

Several limitations are worth bearing in mind when interpreting the results of our study. First, OP was reported by parents and proxy reports are vulnerable to social desirability and recall bias. For example, parents may have been unaware of the time that their child played outdoors at school, resulting in measurement error. Second, despite efforts to recruit a national sample, there may have been self-selection bias and there was a high dropout rate, but effect sizes for most differences between dropouts and completers were generally trivial, reducing concerns about external validity. Third, averaging data for weather variables across the week preceding the survey may have weakened associations with OP, but this was necessary to align with the OP recall period. Fourth, given the number of potential correlates explored, there is a considerable risk of type I error, though we noted that many of the significant correlates were at the $p < 0.001$ threshold in the multivariable model. Fifth, the use of postal codes instead of home addresses (for privacy reasons) limits the precision of built environment measures, especially in rural areas where postal codes represent a larger geographical area. Sixth, some constructs such as traffic safety had low internal consistency, but were retained in the analysis based on systematic review evidence of association with OP (Lambert et al., 2019). Seventh, it is possible that the correlates of OP vary by characteristics such as child age and differ between weekdays and weekend days (Dahl et al., 2024), underscoring a need for future research. Eighth, we could not explore the associations between the policy level of the social-ecological model (Sallis et al., 2006) and OP because we did not collect such data. This level has been understudied in relation to OP (Larouche et al., 2023a) and is worthy of more investigation. Last, it is difficult to disentangle the potential influence of study wave from that of COVID-19 restrictions, which decreased through the study period. In contrast, the national longitudinal design, inclusion of four study waves, large sample at baseline, inclusion of correlates of OP representing multiple levels of evidence of the social-ecological model, and moderation analyses are important study strengths.

5. Conclusion

We found that the correlates of OP during the COVID-19 pandemic spanned multiple levels of influence of the social-ecological model, suggesting that interventions to promote OP should target multiple

levels. Moreover, child gender moderated the associations between five correlates of OP, underscoring a need for future research exploring gender as a moderator. Future interventions should target modifiable correlates of OP such as dog and phone ownership, IM, parents' perceived behavioral control, and social cohesion. Additional efforts may be needed to prevent the age-related decline in children's OP and encourage OP during colder seasons. Consistent messages about the relative safety of outdoor versus indoor environments should be provided and efforts should be made to preserve opportunities for children to play outdoors with peers.

CRediT authorship contribution statement

Richard Larouche: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Robert T. Duffy:** Writing – review & editing, Formal analysis. **Kristian Larsen:** Writing – review & editing, Methodology, Formal analysis. **Mathieu Bélanger:** Writing – review & editing, Methodology, Funding acquisition. **Mariana Brussoni:** Writing – review & editing, Methodology, Funding acquisition. **Guy Faulkner:** Writing – review & editing, Methodology, Funding acquisition. **Katie Gunnell:** Writing – review & editing, Methodology, Funding acquisition. **Mark S. Tremblay:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Data statement

The data are not publicly available because participants have not consented to this, but they are available from the corresponding author pending completion and approval of a data-sharing agreement.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Richard Larouche reports financial support was provided by Heart and Stroke Foundation of Canada. RL receives royalties from Elsevier for his book, *Children's Active Transportation*. Other authors have no competing interests to declare. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was funded by a grant-in-aid from the Heart and Stroke Foundation of Canada (grant # G-19-0026216). The funder had no role in study design, data collection, analysis, interpretation of data, writing of the report, nor the decision to submit the article for publication. We thank Dr. Sarah Moore and Ms. Victoria Hecker for their input in the development of the questionnaire. We thank Mr. Taylor Hecker who developed the Python script that we used to match participants' postal codes with the appropriate dissemination area to link weather data to our dataset. We also thank Dr. Oguchukwu K. Onyeso for his assistance with data treatment and Ms. Jenessa Doctor for creating the graphical abstract. RL holds a Board of Governors Research Chair from the University of Lethbridge.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2025.123625>.

Data availability

The data are not publicly available because participants have not

consented to this, but they are available from the corresponding author pending completion and approval of a data-sharing agreement.

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